

MUHAMMAD BASSIOUNI

SOFTWARE ENGINEER

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SUMMARY

Software Engineer and **Open Source Maintainer** of **LLVM C Library** with 3 years of experience in compilers, toolchain technologies and backend development. Worked at Compiler Research (improving **Clang** compiler support), Menoufia University (enabling digital transformation), Udacity (mentoring educational sessions). Mentored 1 **GSoC** student replacing compiler-rt builtins with LLVM libc. **AWS** Certified Cloud Practitioner. Interested in working on cutting-edge, scalable services technology with global impact.

EXPERIENCE

Compiler Engineer Fellow, Compiler Research, *Remote*.

May 2026 – Nov 2026

A remote, mentor-driven fellowship focusing on high-impact upstream contributions to the LLVM project.

- Improved standard support in Clang compiler by implementing `if` declarations of C2y.
- Implemented C++29 defaulted postfix increment and decrement operators in Clang.

Mentor, Google Summer of Code in LLVM, *Remote*.

Feb 2026 – Aug 2026

Google Summer of Code is a global, online program focused on bringing new contributors into open source development.

- Mentored **one** student for replacing `compiler-rt` floating point builtins with LLVM `libc` math routines.

Backend Software Engineer, Menoufia University, *Menoufia, Egypt*.

Sep 2023 – Aug 2024

Public Egyptian institution providing education, research, and community services across diverse disciplines.

- Reduced yearly subscription costs by **\$40K** by rebuilding outsourcing portal backend with Nest.js & SQL Server.
- Increased system performance by **35%** via migrating to new server host & tech stack using Docker & Node.js.
- Secured API for **88K** users by implementing authorization system using Passport.js & JWT.

Mentor, Udacity, *Menoufia, Egypt*.

Oct 2022 – Dec 2023

Online learning platform offering industry-focused tech and business programs.

- Led & coordinated efforts of **50+** DECI sessions, ensuring alignment with educational goals.
- Delivered **95%** accurate & effective technical content by providing detailed feedback maintaining high standards.
- Collaborated with **10+** instructors & supervisors to enhance learning & create environment for student success.

Backend Software Engineer Intern, Banque Misr, *New Cairo, Egypt*.

Jul 2023 – Sep 2023

Largest bank in Egypt providing retail, corporate, and investment banking services.

- Built real-time currency conversion API using Spring Boot, Docker, and Render for instant conversions.
- Doubled retrieval speed via Redis caching and cron jobs, ensuring reliability with Javadoc, Swagger & JUnit.

OPEN SOURCE PROJECTS

Maintainer, LLVM.

Jan 2025 – Present

LLVM Project is a collection of modular and reusable compiler and toolchain technologies.

[Pull Requests](#)

- **Partnered with Google engineers** to build `libc++` streams by implementing `wctype` in `libc`. [RFC](#)
- **Partnered with Arm engineers** to add C11 bounds-checking interfaces in LLVM `libc`. [RFC](#)
- Implemented C++26 `constexpr` math functions by refactoring **493** math functions to header-only with constant evaluation support in LLVM `libc`. [RFC](#)

- Extended C2y support in Clang by implementing *if declarations* paper. PR
- Implemented *Defaulting Postfix Increment and Decrement Operations* paper of C++29 in Clang. PR
- Improved build accuracy and consistency by fixing LLVM `libc` header generation. PR
- Extended Clang capabilities by adding `_bf16` type support macros with number literal suffix. PR
- Increased test isolation by making `str_to_float_comparison.test` hermetic in LLVM `libc`. PR
- Optimized LLVM IR performance by fixing missed `i128` split optimization. PR

SKILLS

Programming Languages	Python, TypeScript, C, C++, Java
Frameworks & Libraries	React Native, Flask, Nest.js, Express.js, Spring Boot
Databases	MySQL, PostgreSQL, Redis
Tools	Node.js, Git, Graphite, Docker, Bash, Linux, Postman, AWS
Languages	English (Advanced), Arabic (Native)

EDUCATION

Bachelor of Artificial Intelligence in Intelligent Systems, Menoufia University, *Egypt*. Oct 2021 – Jun 2025

PROJECTS

GitHub Link. github.com/bassiounix/open-source-projects

Lox Interpreter. Educational interpreter in Java & C; Built compiler with custom tokenizer, parser, AST generation, static analysis, and interpreter for context-aware execution.

Secure HTTP Client. Built end-to-end encrypted HTTP system using C++ & `httplib`. Utilized RC4 encryption over network connection & Elliptic-curve Diffie-Hellman key exchange algorithm.

Rosetta. Ancient corpus translator; Facilitated ancient hieroglyph translation by training AI models with PyTorch and YOLOv8, and built a React Native app integrating detection, translation, and personal history features.

Wind. Developed reproducible environment configurations on Windows 11 using PowerShell 7, `winget` & `vcpkg`.

Zoi. AI-driven Tic-Tac-Toe; Built tic-tac-toe game using Rust and Min-max algorithm.

CERTIFICATES & AWARDS

GEM Hackathon 2025 Third Edition, by The Grand Egyptian Museum & Legacy. Dec 2025

AWS Certified Cloud Practitioner, foundational certification in AWS Cloud. Nov 2024

FinYology FinTech Hackathon, 4th place finalist among 50+ teams – Central Bank of Egypt. Oct 2023

MINT Fintech Hackathon, 3rd place startup idea among 10 teams – EGBank & Startup Factory. Aug 2023

Smart Governance Hackathon, 4th place in Money Governance category – AWS & Benha University. Feb 2023

NTI AI Summer Training, Training in Python, Machine Learning, Deep Learning & AI Deployment. Oct 2022

Full-Stack Web Development Nanodegree, EgFWD scholarship – Udacity & MCIT. May 2022

EXTRA-CURRICULAR ACTIVITIES

- Created tech-oriented YouTube channel sharing experiences.
- Joined Students for Egypt Chapter as Vice Head of Development Team and gave several seminars.
- Founded Community-X (Phoenix) Chapter and worked as events facilitator.